

3.1 The Inner Product

3.1.1 Definition

In Note 1, we introduced vector spaces and studied their structure through bases, span, and linear independence. We now introduce the inner product – a generalization of the dot product that allows us to multiply two vectors together and quantify lengths and angles.

Definition 3.1 (Inner Product): Given a complex vector space $\mathbb{V}$, an **inner product** is a function $\langle \cdot, \cdot \rangle : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{C}$ that takes two vectors and returns a scalar, satisfying the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{V}$ and all $\alpha, \beta \in \mathbb{C}$:

1. **Linearity:** $\langle \alpha \mathbf{u} + \beta \mathbf{w}, \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle + \beta \langle \mathbf{w}, \mathbf{v} \rangle$.
2. **Conjugate symmetry:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle^*$.
3. **Positive semi-definiteness:** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$.

Throughout this note, z^* denotes the complex conjugate of a scalar $z \in \mathbb{C}$, and $\mathbf{v}^*$ denotes the vector obtained by taking the complex conjugate of each entry of $\mathbf{v}$.

Note that when the vectors are real-valued, conjugate symmetry reduces to ordinary symmetry: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.

Definition 3.2 (Standard Inner Product): The standard inner product on $\mathbb{C}^n$ is defined as:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T \mathbf{w}^* = \sum_{i=1}^n v_i w_i^* \quad (1)$$

For real vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, this simplifies to the dot product:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T \mathbf{w} = \sum_{i=1}^n v_i w_i \quad (2)$$

Unless stated otherwise, when we write $\langle \mathbf{v}, \mathbf{w} \rangle$ for any two vectors $\mathbf{v}, \mathbf{w}$, we mean the standard inner product.

Example 3.1 (Verifying the inner product axioms for the standard inner product):

We verify that the standard inner product on $\mathbb{C}^n$ satisfies all three inner product axioms. Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and $\alpha, \beta \in \mathbb{C}$.

1. Linearity: We want to show that $\langle \alpha \mathbf{u} + \beta \mathbf{w}, \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle + \beta \langle \mathbf{w}, \mathbf{v} \rangle$.

$$\begin{aligned} \langle \alpha \mathbf{u} + \beta \mathbf{w}, \mathbf{v} \rangle &= \sum_{i=1}^n (\alpha u_i + \beta w_i) v_i^* \\ &= \sum_{i=1}^n \alpha u_i v_i^* + \sum_{i=1}^n \beta w_i v_i^* \\ &= \alpha \sum_{i=1}^n u_i v_i^* + \beta \sum_{i=1}^n w_i v_i^* \\ &= \alpha \langle \mathbf{u}, \mathbf{v} \rangle + \beta \langle \mathbf{w}, \mathbf{v} \rangle \end{aligned}$$

2. Conjugate symmetry: We want to show that $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle^*$.

$$\begin{aligned} \langle \mathbf{v}, \mathbf{u} \rangle^* &= \left(\sum_{i=1}^n v_i u_i^* \right)^* \\ &= \sum_{i=1}^n (v_i u_i^*)^* \\ &= \sum_{i=1}^n v_i^* u_i \\ &= \sum_{i=1}^n u_i v_i^* \\ &= \langle \mathbf{u}, \mathbf{v} \rangle \end{aligned}$$

3. Positive semi-definiteness: We want to show that $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ with equality if and only if $\mathbf{v} = \mathbf{0}$.

$$\begin{aligned} \langle \mathbf{v}, \mathbf{v} \rangle &= \sum_{i=1}^n v_i v_i^* \\ &= \sum_{i=1}^n |v_i|^2 \end{aligned}$$

Since $|v_i|^2 \geq 0$ for all i , the sum is non-negative and real. Furthermore, $\sum_{i=1}^n |v_i|^2 = 0$ if and only if $v_i = 0$ for all i , that is, $\mathbf{v} = \mathbf{0}$.

Since all three axioms are satisfied, the standard inner product is valid on $\mathbb{C}^n$.

3.1.2 Geometric Interpretation

The inner product has a natural geometric interpretation. For the standard dot product on $\mathbb{R}^n$, we have the relationship

$$\langle \mathbf{v}, \mathbf{w} \rangle = \|\mathbf{v}\| \|\mathbf{w}\| \cos \theta \quad (3)$$

where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$, and $\|\mathbf{v}\|$ denotes the length (norm) of $\mathbf{v}$, which we will define formally in the next section.

This formula tells us that the inner product captures directional information about two vectors:

- If $\langle \mathbf{v}, \mathbf{w} \rangle > 0$, the vectors point in “roughly the same direction” ($\theta < 90^\circ$).
- If $\langle \mathbf{v}, \mathbf{w} \rangle < 0$, the vectors point in “roughly opposite directions” ($\theta > 90^\circ$).
- If $\langle \mathbf{v}, \mathbf{w} \rangle = 0$, the vectors are **orthogonal** (perpendicular, $\theta = 90^\circ$).

These three cases are illustrated in Figure 1.

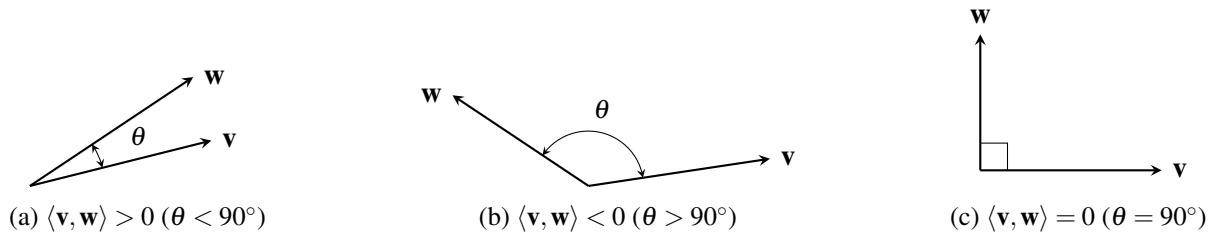


Figure 1: The sign of the inner product reflects the angle between two vectors. (a) Acute angle: vectors point in roughly the same direction. (b) Obtuse angle: vectors point in roughly opposite directions. (c) Right angle: vectors are orthogonal.

3.2 The Norm

3.2.1 Definition

We now formalize the notion of the “length” of a vector.

Definition 3.3 (Norm): Given a vector space $\mathbb{V}$, a **norm** is a function $\|\cdot\| : \mathbb{V} \rightarrow \mathbb{R}$ that assigns a non-negative real number to each vector, satisfying the following properties for all $\mathbf{u}, \mathbf{v} \in \mathbb{V}$ and all $\alpha \in \mathbb{C}$:

1. **Positive semi-definiteness:** $\|\mathbf{v}\| \geq 0$, with $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
2. **Non-negative scaling (homogeneity):** $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$.
3. **Triangle inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

3.2.2 The p-Norm

A widely used family of norms on $\mathbb{R}^n$ is the **p-norm**, defined for $p \geq 1$:

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p} \quad (4)$$

Some important special cases:

- **ℓ_1 norm** ($p = 1$): $\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|$. This is the sum of the magnitudes of each component.
- **ℓ_2 norm** ($p = 2$): $\|\mathbf{v}\|_2 = \sqrt{\sum_{i=1}^n |v_i|^2}$.
- **ℓ_∞ norm** ($p \rightarrow \infty$): $\|\mathbf{v}\|_\infty = \max_{1 \leq i \leq n} |v_i|$. This is the magnitude of the highest-magnitude component.

3.2.3 The Euclidean Norm

Perhaps the most commonly used norm is the **Euclidean norm** (ℓ_2 norm):

$$\|\mathbf{v}\|_2 = \sqrt{|v_1|^2 + |v_2|^2 + \cdots + |v_n|^2} = \sqrt{\sum_{i=1}^n |v_i|^2} \quad (5)$$

Observe that this can be written directly in terms of the standard inner product:

$$\|\mathbf{v}\|_2 = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} \quad (6)$$

This connection lets us verify the norm axioms using inner product properties. Unless otherwise specified, from here on $\|\mathbf{v}\|$ (with no subscript) refers to the Euclidean norm.

Positive semi-definiteness: By positive semi-definiteness of the inner product, $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, so $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} \geq 0$. Furthermore, $\|\mathbf{v}\| = 0$ if and only if $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, which by positive semi-definiteness holds if and only if $\mathbf{v} = \mathbf{0}$.

Non-negative scaling (homogeneity):

$$\begin{aligned}\|\alpha \mathbf{v}\| &= \sqrt{\langle \alpha \mathbf{v}, \alpha \mathbf{v} \rangle} \\ &= \sqrt{\alpha \alpha^* \langle \mathbf{v}, \mathbf{v} \rangle} && \text{(by linearity and conjugate symmetry)} \\ &= \sqrt{|\alpha|^2 \langle \mathbf{v}, \mathbf{v} \rangle} \\ &= |\alpha| \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} \\ &= |\alpha| \|\mathbf{v}\|\end{aligned}$$

Normalization: A useful consequence of non-negative scaling is that any nonzero vector can be rescaled to have length one. Given $\mathbf{v} \neq \mathbf{0}$, the **normalized** vector

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad (7)$$

points in the same direction as $\mathbf{v}$ and satisfies $\|\hat{\mathbf{v}}\| = \frac{1}{\|\mathbf{v}\|} \|\mathbf{v}\| = 1$. Such a vector is called a **unit vector**.

The proof of the triangle inequality for the Euclidean norm relies on the Cauchy-Schwarz inequality, which we state at the end of this note.

3.3 Applications

Norms and inner products have many useful applications, including in the Cauchy-Schwarz inequality and the triangle inequality. Both are presented below:

Theorem 3.1: (Cauchy-Schwarz Inequality). For any vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$:

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \cdot \|\mathbf{w}\| \quad (8)$$

with equality if and only if $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent.

Theorem 3.2: (Triangle Inequality). For any vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$:

$$\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\| \quad (9)$$

The proof of both will be left out of these notes for now.